Experiment 7.

Create a Service in React That Fetches Weather Information from OpenWeatherMap.org and Displays the Current and Historical Weather Information Using Graphical Representation Using Chart.js

AIM: To create a service in React that fetches weather data and displays it in a graph.

DESCRIPTION:

To create a React service that fetches weather information from OpenWeatherMap.org and displays both current and historical weather data using Chart.js, you can follow these steps. This guide assumes you have a basic understanding of React and JavaScript.

Step 1: Set Up Your React Application

1. Create a New React App:

Open your terminal and create a new React application using Create React App.

Run the project creation command.

npm create vite@latest weather-app

Project name: weather-app

Select a framework: React

Select a variant: JavaScript

cd weather-app

Start the development server.

2. Install Additional Packages (Optional):

If you plan to use any additional libraries (like Bootstrap for styling), you can install them. For example, to install Bootstrap:

npm i

npm install bootstrap

npm install react-router-dom

npm install axios chart.js react-chartjs-2

Run application use this command **npm run dev**

Step 2: Get Your OpenWeatherMap API Key

1. Sign Up: Go to [OpenWeatherMap](https://openweathermap.org/) and sign up for an account.
2. Get API Key: After signing up, go to the API section and get your API key.

Step 3: Create a Weather Service

Create a new folder named services in the src directory and create a file named WeatherService.js.

App.jsx

```
import React, { useState } from "react";
import axios from "axios";
import { Line } from "react-chartjs-2";
import './App.css';
import { Chart as ChartJS, CategoryScale, LinearScale, PointElement, LineElement, Title, Tooltip, Legend } from
"chart.js";
ChartJS.register(CategoryScale, LinearScale, PointElement, LineElement, Title, Tooltip, Legend);
const API_KEY = "f6dc4f4e7b873301bc6afaeb13e4ea58"; // Replace with your key
const WEATHER_URL = "https://api.openweathermap.org/data/2.5/weather";
const FORECAST_URL = "https://api.openweathermap.org/data/2.5/forecast";
const App = () =>{
  const [city, setCity] = useState("Hyderabad");
  const [weatherData, setWeatherData] = useState(null);
  const fetchWeather = async (cityName) =>{
    try {
      console.log("Fetching weather for:", cityName);
      // Fetch current weather
const currentWeatherResponse = await axios.get(WEATHER_URL, {
  params: { q: cityName, units: "metric", appid: API_KEY },
});
console.log("Current Weather Data:", currentWeatherResponse.data);
// Fetch 5-day forecast
const forecastResponse = await axios.get(FORECAST_URL, {
  params: { q: cityName, units: "metric", appid: API_KEY },
});
console.log("Forecast Data:", forecastResponse.data);
// Extract relevant data for the next 5 days
const forecastData = forecastResponse.data.list.filter((entry, index) =>index % 8 === 0).slice(0, 5);
setWeatherData({
  currentTemp: currentWeatherResponse.data.main.temp,
    forecast: forecastData,
  });
} catch (error) {
  console.error("Error fetching weather:", error);
  alert("Could not fetch weather data. Please check the city name or try again later.");
}
```

```
};
const handleSearch = () => {
  if (!city.trim()) {
    alert("Please enter a city name.");
    return;
  }
  fetchWeather(city);
};
return (
  <section className="py-5">
    <div className="row justify-content-center mt-5">
      <div className="col-md-6">
        <div className="card shadow">
          <div className="card-body">
           <h2>Weather Forecast</h2>
            <div className="d-flex">
             <input
               type="text" className="form-control w-50"
               value={city}
               onChange={(e) => setCity(e.target.value)}
               placeholder="Enter city name"
             />
  <button className="btn btn-primary ms-2 text-white" onClick={handleSearch}>
    Get Weather
  </button>
</div>
{weatherData ? (
  <div className="bg-white p-3 mt-3 rounded shadow">
   <h4>Current Temperature in {city}: {weatherData.currentTemp}°  C</h4>
   <div style={{ width: "400px", height: "250px", margin: "auto" }}>
     <Line
       data={{
         labels: weatherData.forecast.map((d) => new Date(d.dt * 1000).toLocaleDateString()),
         datasets: [
           {
     label: `Temperature Forecast (°  C)`,
     data: weatherData.forecast.map((d) => d.main.temp),
     borderColor: "blue",
     backgroundColor: "rgba(0, 0, 255, 0.3)",
   },
  ],
}}
```

```
        options={{
          maintainAspectRatio: false, // Ensures it respects div size
        }}
      />
    </div>
  </div>
) : (
  <p>Enter a city and click "Get Weather"</p>
)}
            </div>
          </div>
        </div>
      </div>
    </section>
  );
};
export default App;
```

main.jsx

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import App from "./App.jsx";
import "./App.css";
import 'bootstrap/dist/css/bootstrap.min.css'

createRoot(document.getElementById("root")).render(
 <StrictMode>
   <App />
 </StrictMode>
);
```

.env.example

```
# Copy this file to .env and paste your OpenWeatherMap key
VITE_OWM_KEY=d2db27d78dbc261f005fe6fae43008ab;
```

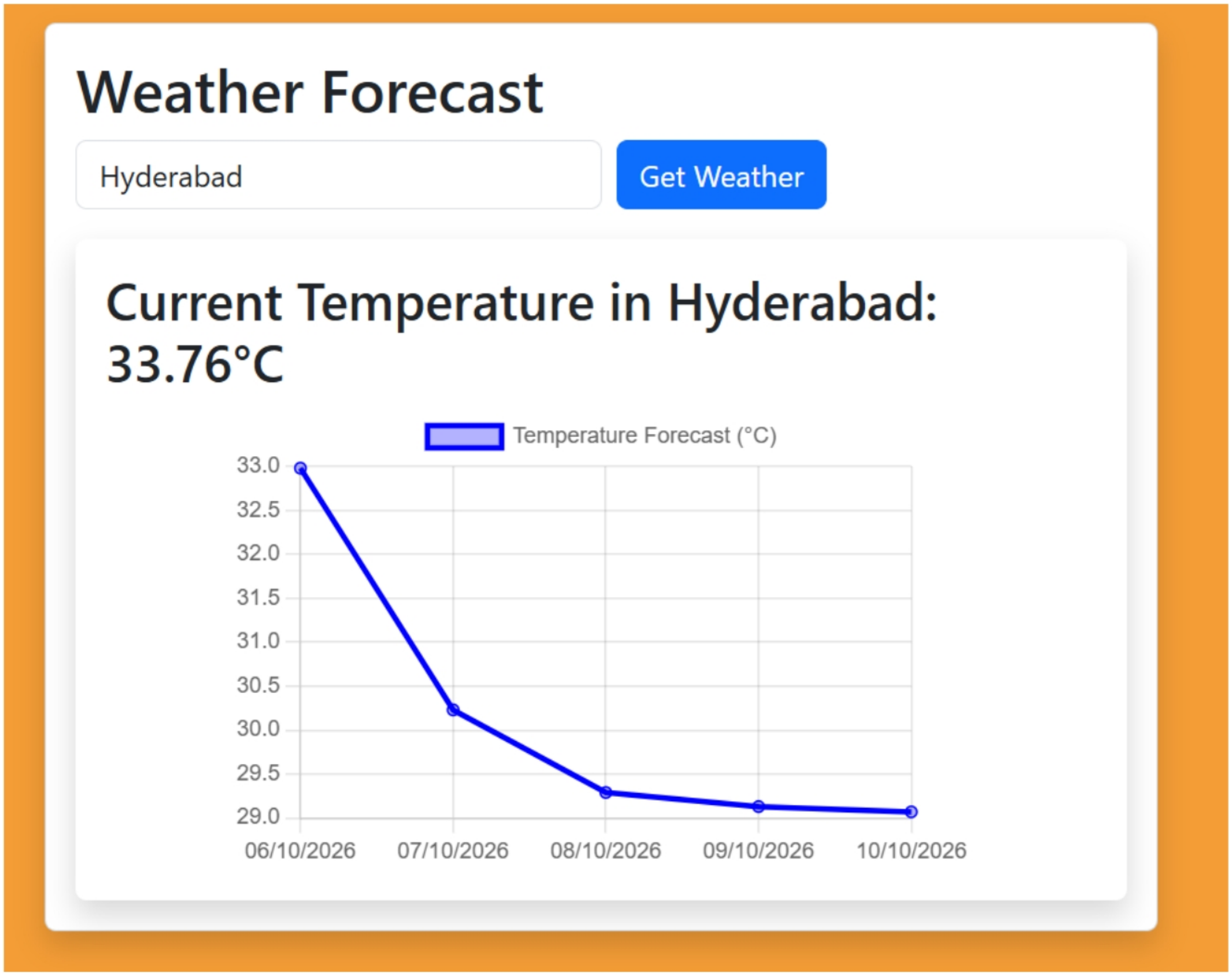
Step 4: Run Your Application

Run your application locally to test it.

npm run dev

Step 5: Access Your Application

Open your browser and go to http://localhost:5173/ .



Viva Questions:

- How would you create a service in React that interacts with the OpenWeatherMap API to fetch weather data?
- What are some key data points you would extract from the OpenWeatherMap API response to display current and historical weather information?
- How would you integrate Chart.js in your React application to visually represent the fetched weather data (e.g., temperature trends)?
- How would you handle error states, such as invalid API keys or network errors, while fetching weather